

Please check the examination details below before entering your candidate information

Candidate surname		Other names	
Centre Number		Candidate Number	

Pearson Edexcel Award

Tuesday 24 June 2025

Afternoon (Time: 3 hours)	Paper reference	9811/01
---------------------------	-----------------	----------------

Advanced Extension Award

Mathematics

Worked solutions

Marked

You must have: Mathematical Formulae and Statistical Tables An insert for questions 5, 6 and 7 (enclosed)	Total Marks
---	-------------

Instructions

- Use **black** ink or ball-point pen.
- **Fill in the boxes** at the top of this page with your name, centre number and candidate number.
- Answer **all** questions.
- Answer the questions in the spaces provided
– *there may be more space than you need.*
- **Calculators may not be used.**
- You must **show all your working.**
- Answers should be given in as simple a form as possible, e.g. $\frac{2\pi}{3}$, $\sqrt{2}$, $3\sqrt{2}$.

Information

- The total mark for this paper is 100 of which **7** marks are for style and clarity of presentation.
- The style and clarity of presentation marks will be indicated as **(+S1)** or **(+S2)**.
- There are 7 questions in this question paper.
- The marks for each question are shown in brackets.
- The total mark for each question is shown at the end of the question.

Advice

- Read each question carefully before you start to answer it.
- Try to answer every question.
- Check your answers if you have time at the end.

Turn over ►

P76378A

©2025 Pearson Education Ltd.
Y:1/1/1/



1. Using the substitution $u = \frac{\pi}{2}\sqrt{1+x}$ evaluate

$$\int_{\frac{5}{4}}^{\frac{21}{4}} \sec^2\left(\frac{\pi}{2}\sqrt{1+x}\right) dx$$

(10)

$$u = \frac{\pi}{2} (1+x)^{1/2} \quad \left(\frac{2u}{\pi}\right)^{-1} (1+x)^{-1/2}$$

$$\frac{du}{dx} = \frac{\pi}{2} \times \frac{1}{2} (1+x)^{-1/2} = \frac{\pi}{4} (1+x)^{-1/2} = \frac{\pi}{4} \times \left(\frac{2u}{\pi}\right)^{-1} = \frac{\pi^2}{8u}$$

$$du = \frac{\pi^2}{8u} dx$$

$$\text{When } x = \frac{21}{4} \quad u = \frac{\pi}{2} \sqrt{\frac{25}{4}} = \frac{5\pi}{4}$$

$$\frac{8u}{\pi^2} du = dx$$

$$\text{When } x = \frac{5}{4} \quad u = \frac{\pi}{2} \sqrt{\frac{9}{4}} = \frac{3\pi}{4}$$

$$\int \sec^2(u) dx = \int \sec^2(u) \cdot \frac{8u}{\pi^2} du = \frac{8}{\pi^2} \int u \sec^2(u) du$$

$$a = u \quad b = \tan(u)$$

$$\frac{da}{du} = 1 \quad \frac{db}{du} = \sec^2(u)$$

$$\frac{8}{\pi^2} \left[u \tan(u) - \int \tan(u) du \right]_{3\pi/4}^{5\pi/4} = \frac{8}{\pi^2} \left[u \tan(u) - \ln|\sec(u)| \right]_{3\pi/4}^{5\pi/4}$$

$$= \frac{8}{\pi^2} \left[\left(\frac{5\pi}{4} \tan\left(\frac{5\pi}{4}\right) - \ln\left|\sec\left(\frac{5\pi}{4}\right)\right| \right) - \left(\frac{3\pi}{4} \tan\left(\frac{3\pi}{4}\right) - \ln\left|\sec\left(\frac{3\pi}{4}\right)\right| \right) \right]$$

$$= \frac{8}{\pi^2} \left[\frac{5\pi}{4} - \ln|-\sqrt{2}| - \left(-\frac{3\pi}{4} + \ln|-\sqrt{2}| \right) \right]$$

$$\frac{8}{\pi^2} \left[\frac{8\pi}{4} \right] = \frac{16}{\pi}$$



Question 1 continued

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA



Question 1 continued

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA



Question 1 continued

(Total for Question 1 is 10 marks)

2. (a) On separate diagrams sketch the graph of

(i) $y = |x^2 - 4|$ (3)

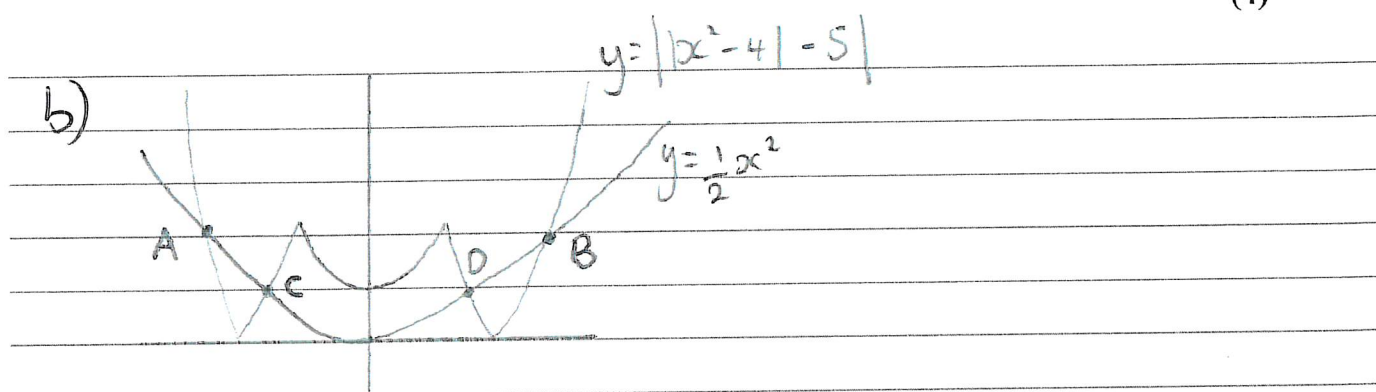
(ii) $y = ||x^2 - 4| - 5|$ (4)

Indicate clearly on each sketch the coordinates of

- any vertices
- any points where the graph touches or crosses the coordinate axes

(b) Hence solve the equation

$$||x^2 - 4| - 5| = \frac{1}{2}x^2 \quad (4)$$



Points A and B are when $\frac{1}{2}x^2$ intersect the original x^2 translated

$$x^2 - 4 - 5 = \frac{1}{2}x^2 \quad M1$$

$$x^2 - 9 = \frac{1}{2}x^2$$

$$\frac{1}{2}x^2 = 9$$

$$x^2 = 18$$

$$x = \pm \sqrt{18} = \pm \sqrt{9 \cdot 2} = \pm 3\sqrt{2} \quad A1$$

Points C and D are when $\frac{1}{2}x^2$ intersect the reflected $x^2 - 4$ translated

$$-[(x^2 - 4) - 5] = \frac{1}{2}x^2 \quad dM1$$

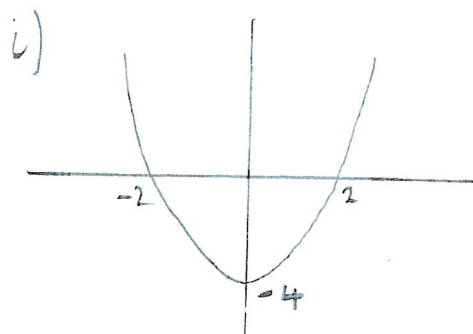
$$-x^2 + 9 = \frac{1}{2}x^2$$

$$9 = \frac{3}{2}x^2$$

$$6 = x^2 \quad x = \pm \sqrt{6} \quad A1$$



Question 2 continued

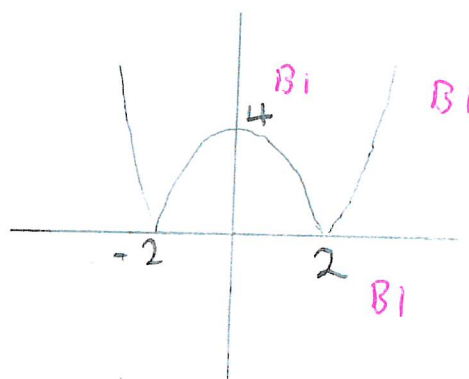


$$y = x^2 - 4$$

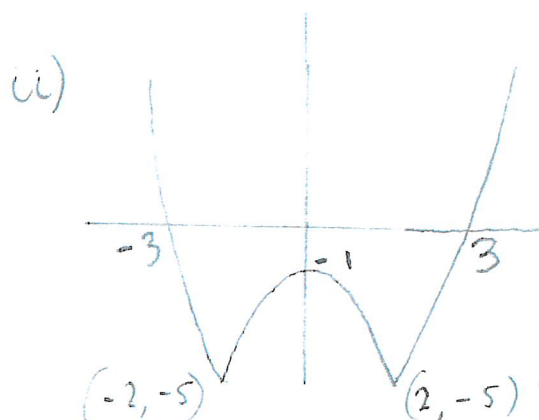
$$0 = x^2 - 4$$

$$4 = x^2$$

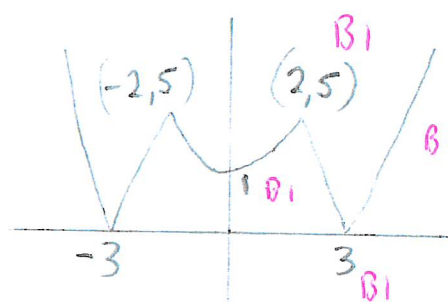
$$\pm 2 = x$$



$$y = |x^2 - 4|$$



$$y = |x^2 - 4| - 5$$



$$y = ||x^2 - 4| - 5|$$

Roots on original curve
not reflected so

$$0 = (x^2 - 4) - 5$$

$$0 = x^2 - 9$$

$$9 = x^2$$

$$\pm 3 = x$$

Question 2 continued

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA



Question 2 continued

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

(Total for Question 2 is 11 marks)

3. Solve the equation

$$\arcsin 2x = \arccos x - \frac{\pi}{6} \quad 0 \leq x \leq 0.5$$

(5)

$$\sin(\arcsin(2x)) = \sin(\arccos(x) - \pi/6)$$

$$2x = \sin \arccos(x) \cos(\pi/6) - \sin(\pi/6) \cos \arccos(x) \quad M1$$

$$2x = \sin \arccos(x) \cdot \frac{\sqrt{3}}{2} - \frac{1}{2}x$$

$$\frac{5x}{2} = \frac{\sqrt{3}}{2} \sin \arccos(x)$$

$$\frac{5x}{\sqrt{3}} = \sin \arccos(x)$$

$$\sin^2 \theta + \cos^2 \theta = 1$$

$$\sin^2 \theta = 1 - \cos^2 \theta$$

$$\sin \theta = \sqrt{1 - \cos^2 \theta}$$

$$\theta = \arccos(x)$$

$$\sin \arccos(x) = \sqrt{1 - (\cos \arccos(x))^2} = \sqrt{1 - x^2}$$

$$\left(\frac{5x}{\sqrt{3}}\right)^2 = \left(\sqrt{1 - x^2}\right)^2 \quad M1 \quad A1$$

$$\frac{25x^2}{3} = 1 - x^2$$

$$\frac{28x^2}{3} = 1$$

$$x^2 = \frac{3}{28}$$

$$x = \frac{\sqrt{3}}{\sqrt{28}} = \frac{\sqrt{3}}{\sqrt{28}} = \frac{\sqrt{3}}{2\sqrt{7}} = \frac{\sqrt{3} \times \sqrt{7}}{2\sqrt{7} \times \sqrt{7}} = \frac{\sqrt{21}}{14}$$

dd M1

A1



Question 3 continued

(Total for Question 3 is 5 marks)

4. In this question you may assume that $\sqrt{p}$ is irrational for any prime number p .

- (a) Use proof by contradiction to show that there is no rational number a and integer b for which

$$\sqrt{5} = a + b\sqrt{7} \quad (5)$$

- (b) Hence prove that $\sqrt{5} + \sqrt{7}$ is not a rational number.

(2)
(+S1)

Assume there is a rational number a and integer b such that $\sqrt{5} = a + b\sqrt{7}$ B1

$$(\sqrt{5} - a)^2 = (b\sqrt{7})^2$$

$$5 - 2a\sqrt{5} + a^2 = 7b^2 \quad \text{M1}$$

$$5 + a^2 - 7b^2 = 2a\sqrt{5}$$

If a is rational a^2 is rational

If b is an integer b^2 is rational

So $a^2 - 7b^2 + 5$ is rational M1

This is a contradiction as $2a\sqrt{5}$ is irrational unless $a = 0$

(S*) If $a = 0$ $5 = 7b^2$ so $5/7 = b^2$ which is not possible as b is an integer so $a \neq 0$ B1
therefore contradiction.

Hence there is no rational a and integer b for which $\sqrt{5} = a + b\sqrt{7}$ A1

b) If $b = -1$ then from (a) $\sqrt{5} = a - \sqrt{7}$ and a must be irrational. M1

$\sqrt{5} + \sqrt{7} = a$
therefore $\sqrt{5} + \sqrt{7}$ is irrational A1



Question 4 continued

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

Handwriting practice area with 30 horizontal lines.



Question 4 continued

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA



Question 4 continued

(Total for Question 4 is 8 marks)

5. The discrete random variable X has the probability distribution

n	2	4	6	8	10
$P(X=n)$	$\frac{1}{10} \operatorname{cosec}^2 x$	a	$\frac{1}{5} \cot^2 x$	$\frac{1}{10} \sec^2 x$	$\frac{1}{5} \tan^2 x$

where a and x are constants.

(a) Show that

$$\sin^2 2x = \frac{6}{7-5a} \quad (7)$$

(b) Use the result from part (a) to find a restriction on the maximum possible value of a . (2)

The random variables X_1 and X_2 are independent and each have the same distribution as X .

Given that $P(X_1 \times X_2 = 16) = a - \frac{1}{25}$

(c) determine any possible values of a , justifying your answer. (8)
(+S2)

$$\sum P(X=n) = 1$$

$$\frac{1}{10} \operatorname{cosec}^2(x) + a + \frac{1}{5} \cot^2(x) + \frac{1}{10} \sec^2(x) + \frac{1}{5} \tan^2(x) = 1 \quad B1$$

$$\textcircled{\times 10} \quad \frac{1}{\sin^2(x)} + 10a + 2 \frac{\cos^2(x)}{\sin^2(x)} + \frac{1}{\cos^2(x)} + 2 \frac{\sin^2(x)}{\cos^2(x)} = 10 \quad M1$$

$$\frac{1 + 2 \cos^2(x)}{\sin^2(x)} + \frac{2 \sin^2(x) + 1}{\cos^2(x)} = 10 - 10a$$

$$\textcircled{\times \sin^2(x) \cos^2(x)} \quad (1 + 2 \cos^2(x)) \cos^2(x) + (2 \sin^2(x) + 1) \sin^2(x) = (10 - 10a) \cos^2(x) \sin^2(x) \quad M1$$

$$\cos^2(x) + 2 \cos^4(x) + 2 \sin^4(x) + \sin^2(x) = (10 - 10a) \cos^2(x) \sin^2(x)$$

$$2(\cos^4(x) + \sin^4(x)) = 2[(\cos^2(x) + \sin^2(x))^2 - 2 \sin^2(x) \cos^2(x)]$$

$$\cos^2(x) + \sin^2(x) = 1 \quad = 2[1 - 2 \sin^2(x) \cos^2(x)] = 2 - 4 \sin^2(x) \cos^2(x)$$

$$\sin(2x) = 2 \sin(x) \cos(x) \quad \sin^2(2x) = 4 \sin^2(x) \cos^2(x)$$



Question 5 continued

$$\cos^2(x) + \sin^2(x) + 2(\cos^4(x) + \sin^4(x)) = (10 - 10a)\cos^2(x)\sin^2(x)$$

$$1 + 2 - 4\sin^2(x)\cos^2(x) = \left(\frac{10 - 10a}{4}\right) \cdot 4\cos^2(x)\sin^2(x) \quad M1$$

$$3 - \sin^2(2x) = \left(\frac{10 - 10a}{4}\right) \sin^2(2x) \quad dM1$$

$$(x4) \quad 12 - 4\sin^2(2x) = (10 - 10a)\sin^2(2x)$$

$$12 = (14 - 10a)\sin^2(2x) \quad M1$$

$$+ 4\sin^2(2x)$$

$$\frac{12}{14 - 10a} = \frac{6}{7 - 5a} = \sin^2(2x) \quad A1$$

b) Max $\sin^2(2x) = 1$

$$\frac{6}{7 - 5a} = 1 \quad M1 \quad 6 = 7 - 5a$$

$$5a = 1$$

$$a = 1/5 \quad A1$$

c) Factor pairs of 16

1, 16

2, 8 ✓

4, 4 ✓

8, 2 ✓

16, 1

$$2 \times \frac{1}{10} \operatorname{cosec}^2(x) \times \frac{1}{10} \sec^2(x) + a^2 = a - \frac{1}{25} \quad M1 \quad \text{RHS not needed}$$

$$\frac{2}{100 \sin^2(x) \cos^2(x)} + a^2 = a - \frac{1}{25}$$

$$\frac{2}{25 \sin^2(2x)} + a^2 = a - \frac{1}{25} \quad M1 \quad A1 \quad \text{RHS not needed} \quad (x25)$$

$$\frac{2}{\sin^2(2x)} + 25a^2 = 25a - 1$$

$$\frac{2}{6} = 2 \times \frac{7 - 5a}{6} = \frac{7 - 5a}{3}$$

$$7 - 5a$$

$$\frac{7 - 5a}{3} + 25a^2 = 25a - 1 \quad M1 \quad (x3)$$

$$7 - 5a + 75a^2 = 75a - 3$$

$$75a^2 - 80a + 10 = 0 \quad A1 \quad (x5)$$

$$15a^2 - 16a + 2 = 0$$

Question 5 continued

$$a = \frac{16 \pm \sqrt{16^2 - 4 \times 15 \times 2}}{2 \times 15} = \frac{16 \pm \sqrt{256 - 120}}{30}$$

$$a = \frac{16 \pm \sqrt{136}}{30} = \frac{16 \pm 2\sqrt{34}}{30} = \frac{8 \pm \sqrt{34}}{15}$$

$$\begin{array}{ll} \sqrt{34} > \sqrt{25} & \sqrt{34} > 5 \\ \sqrt{34} < \sqrt{36} & \sqrt{34} < 6 \end{array}$$

$$\frac{8 + \sqrt{34}}{15} > \frac{8 + 5}{15} = \frac{13}{15} > \frac{3}{5} = \frac{1}{5} \quad \text{We know from (b) max } a = \frac{1}{5} \text{ not possible.}$$

$$\frac{8 - \sqrt{34}}{15} < \frac{8 - 6}{15} = \frac{2}{15} < \frac{3}{15} = \frac{1}{5}$$

$$\text{So } a = \frac{8 - \sqrt{34}}{15} \quad \text{A1}$$



Question 5 continued

(Total for Question 5 is 19 marks)

6. Relative to a fixed origin O , the points A , B and C have position vectors

$$\vec{OA} = 10\mathbf{i} + 5\mathbf{j} + 4\mathbf{k} \quad \vec{OB} = -4\mathbf{i} + 7\mathbf{j} - \mathbf{k} \quad \vec{OC} = -2\mathbf{i} + 17\mathbf{j} + 10\mathbf{k}$$

- (a) Show that triangle ABC is isosceles.

(2)

The circle S lies inside triangle ABC such that each side of the triangle is a tangent to S .

The centre, X , of S lies on the angle bisector of each vertex of the triangle.

The point D lies on AC such that $|\vec{AD}| = |\vec{AB}|$

- (b) Find the position vector of D .

(2)

- (c) Find the position vector of the midpoint of BD .

(2)

- (d) Hence show that the position of X is given by

$$\vec{OX} = 10\mathbf{i} + 5\mathbf{j} + 4\mathbf{k} + p(-2\mathbf{i} + \mathbf{j})$$

where p is a scalar constant.

(4)

- (e) Use the angle bisector of vertex B to find another expression for $\vec{OX}$ in terms of a scalar constant q .

(3)

- (f) By equating the $\mathbf{k}$ components of the two expressions for $\vec{OX}$, determine the value of q .

(2)

- (g) Find

- (i) the position vector of the centre of S ,

(2)

- (ii) the radius of S .

(4)

a) $\vec{AB} = \begin{pmatrix} 10 \\ 5 \\ 4 \end{pmatrix} \rightarrow \begin{pmatrix} -4 \\ 7 \\ -1 \end{pmatrix} = \begin{pmatrix} -14 \\ 2 \\ -5 \end{pmatrix}$ $|\vec{AB}| = \sqrt{14^2 + 2^2 + 5^2} = \sqrt{225} = 15$ M1 (+S2)

$\vec{BC} = \begin{pmatrix} -4 \\ 7 \\ -1 \end{pmatrix} \rightarrow \begin{pmatrix} -2 \\ 17 \\ 10 \end{pmatrix} = \begin{pmatrix} 2 \\ 10 \\ 11 \end{pmatrix}$ $|\vec{BC}| = \sqrt{2^2 + 10^2 + 11^2} = \sqrt{225} = 15$

$\vec{AC} = \begin{pmatrix} 10 \\ 5 \\ 4 \end{pmatrix} \rightarrow \begin{pmatrix} -2 \\ 17 \\ 10 \end{pmatrix} = \begin{pmatrix} -12 \\ 12 \\ 6 \end{pmatrix}$ $|\vec{AC}| = \sqrt{12^2 + 12^2 + 6^2} = \sqrt{324} = 18$

$|\vec{AB}| = |\vec{BC}| \therefore \text{isosceles}$ A1



Question 6 continued

$$b) \hat{AC} = \frac{1}{18} \begin{pmatrix} -12 \\ 12 \\ 6 \end{pmatrix}$$

$$\vec{AD} = 15 \hat{AC}$$

$$\vec{OD} = \vec{OA} + \vec{AD} = \begin{pmatrix} 10 \\ 5 \\ 4 \end{pmatrix} + 15 \left(\frac{1}{18} \begin{pmatrix} -12 \\ 12 \\ 6 \end{pmatrix} \right) \quad M_1$$

$$= \begin{pmatrix} 10 \\ 5 \\ 4 \end{pmatrix} + \begin{pmatrix} -10 \\ 10 \\ 5 \end{pmatrix} = \begin{pmatrix} 0 \\ 15 \\ 9 \end{pmatrix} \quad A_1$$

$$c) \vec{BD} = \begin{pmatrix} -4 \\ 7 \\ -1 \end{pmatrix} \rightarrow \begin{pmatrix} 0 \\ 15 \\ 9 \end{pmatrix} = \begin{pmatrix} 4 \\ 8 \\ 10 \end{pmatrix} \quad \frac{1}{2} \vec{BD} = \begin{pmatrix} 2 \\ 4 \\ 5 \end{pmatrix}$$

call the midpoint of BD, M.

$$\vec{OM} = \vec{OB} + \frac{1}{2} \vec{BD} = \begin{pmatrix} -4 \\ 7 \\ -1 \end{pmatrix} + \begin{pmatrix} 2 \\ 4 \\ 5 \end{pmatrix} = \begin{pmatrix} -2 \\ 11 \\ 4 \end{pmatrix} \quad M_1 \quad A_1$$

d) ABD is isosceles so the bisector of angle A goes through the midpoint of BD, M.

$$\vec{AM} = \begin{pmatrix} 10 \\ 5 \\ 4 \end{pmatrix} \rightarrow \begin{pmatrix} -2 \\ 11 \\ 4 \end{pmatrix} = \begin{pmatrix} -12 \\ 6 \\ 0 \end{pmatrix} \quad M_1 \quad A_1$$

X lies on the bisector of angle A. $\vec{AX} = \lambda \begin{pmatrix} -12 \\ 6 \\ 0 \end{pmatrix}$

$$\vec{OX} = \vec{OA} + \vec{AX} = \begin{pmatrix} 10 \\ 5 \\ 4 \end{pmatrix} + \lambda \begin{pmatrix} -12 \\ 6 \\ 0 \end{pmatrix} = \begin{pmatrix} 10 \\ 5 \\ 4 \end{pmatrix} + 6\lambda \begin{pmatrix} -2 \\ 1 \\ 0 \end{pmatrix} \quad M_1$$

$$\vec{OX} = 10\mathbf{i} + 5\mathbf{j} + 4\mathbf{k} + p(-2\mathbf{i} + \mathbf{j}) \quad A_1^* \quad \text{where } p = 6\lambda \quad (5^+)$$

Question 6 continued

e) ABC is isosceles so the bisector of B goes through the midpoint of AC, say N.

$$\vec{ON} = \vec{OA} + \frac{1}{2} \vec{AC} = \begin{pmatrix} 10 \\ 5 \\ 4 \end{pmatrix} + \frac{1}{2} \begin{pmatrix} -12 \\ 12 \\ 6 \end{pmatrix} = \begin{pmatrix} 4 \\ 11 \\ 7 \end{pmatrix} \quad \text{BI}$$

X lies on the bisector of B

$$\vec{BN} = \begin{pmatrix} -4 \\ 7 \\ -1 \end{pmatrix} \rightarrow \begin{pmatrix} 4 \\ 11 \\ 7 \end{pmatrix} = \begin{pmatrix} 8 \\ 4 \\ 8 \end{pmatrix} \quad \vec{BX} = \lambda \begin{pmatrix} 8 \\ 4 \\ 8 \end{pmatrix}$$

$$\vec{OX} = \vec{OB} + \vec{BX} = \begin{pmatrix} -4 \\ 7 \\ -1 \end{pmatrix} + \lambda \begin{pmatrix} 8 \\ 4 \\ 8 \end{pmatrix} = \begin{pmatrix} -4 \\ 7 \\ -1 \end{pmatrix} + 4\lambda \begin{pmatrix} 2 \\ 1 \\ 2 \end{pmatrix} \quad \text{M1 A1}$$

$$\vec{OX} = \begin{pmatrix} -4 \\ 7 \\ -1 \end{pmatrix} + q \begin{pmatrix} 2 \\ 1 \\ 2 \end{pmatrix} \quad \text{where } q = 4\lambda$$

f) k components can be equated: $4 = -1 + 2q$ M1
 $\frac{5}{2} = q$ A1

$$\text{gi) } \vec{OX} = \begin{pmatrix} -4 \\ 7 \\ -1 \end{pmatrix} + \frac{5}{2} \begin{pmatrix} 2 \\ 1 \\ 2 \end{pmatrix} = \begin{pmatrix} 1 \\ 9.5 \\ 4 \end{pmatrix} \quad \text{M1 A1}$$

gii) N lies on the circle, hence $|\vec{XN}| = r$

$$\vec{XN} = \begin{pmatrix} 1 \\ 9.5 \\ 4 \end{pmatrix} \rightarrow \begin{pmatrix} 4 \\ 11 \\ 7 \end{pmatrix} = \begin{pmatrix} 3 \\ 1.5 \\ 3 \end{pmatrix} \quad |\vec{XN}| = \sqrt{3^2 + 1.5^2 + 3^2} \quad \text{M1 A1}$$

$$= \sqrt{18 + \frac{9}{4}} = \sqrt{\frac{81}{4}}$$

$$r = \frac{9}{2} \quad \text{A1}$$



Question 6 continued

DO NOT WRITE IN THIS AREA



Question 6 continued

Lined area for writing the answer to Question 6.

DO NOT WRITE IN THIS AREA



Question 6 continued

(Total for Question 6 is 23 marks)

7. Given $u = y^x$

- (a) by taking $\ln$ of both sides, or otherwise, determine $\frac{du}{dx}$ in terms of x, y and $\frac{dy}{dx}$ only. (5)

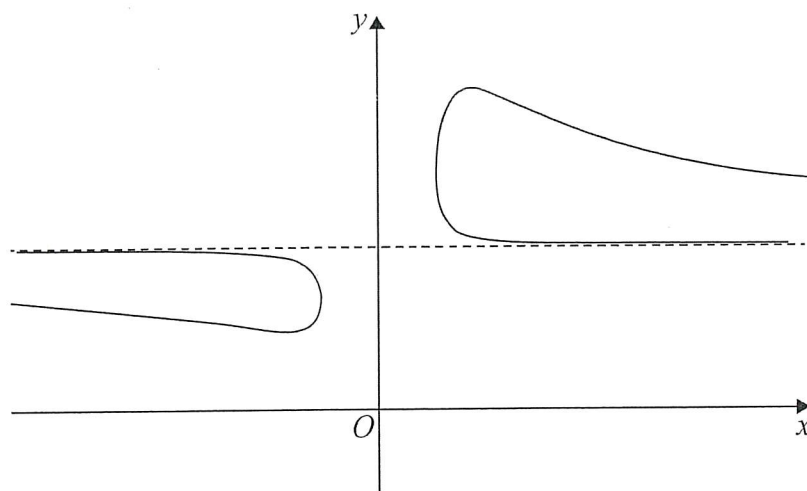


Figure 1

Figure 1 shows a sketch of the curve given by the equation

$$3y^x - x^3 \ln y = 0$$

The curve has a horizontal asymptote $y = 1$, and is defined for

- $x \leq -a, \quad x \geq a$
- $b \leq y \leq c \quad y \neq 1$

where a, b and c are positive constants.

- (b) Determine $\frac{dy}{dx}$ for this curve. (4)
- (c) Find the value of a . (6)
- (d) Find the value of b and the value of c . (7)

(+S2)

$$\begin{aligned} \text{a) } \ln u &= x \ln y \quad \text{B1} \quad \text{RHS} \quad a=x \quad b=\ln y \\ \frac{1}{u} \frac{du}{dx} &= \ln y + \frac{x}{y} \frac{dy}{dx} \quad \text{M1 A1} \quad \frac{da}{dx} = 1 \quad \frac{db}{dx} = \frac{1}{y} \frac{dy}{dx} \\ &\quad \frac{b da}{dx} + a \frac{db}{dx} \\ \frac{du}{dx} &= \left(\ln y + \frac{x}{y} \frac{dy}{dx} \right) u = \left(\ln y + \frac{x}{y} \frac{dy}{dx} \right) y^x \quad \text{A1} \end{aligned}$$



Question 7 continued

b) $3y^x = x^3 \ln y$

RHS

$a = x^3$

$b = \ln y$

$\frac{da}{dx} = 3x^2$

$\frac{db}{dx} = \frac{1}{y} \frac{dy}{dx}$

$3 \left[\left(\frac{x}{y} \frac{dy}{dx} + \ln y \right) y^x \right] = 3x^2 \ln y + \frac{x^3}{y} \frac{dy}{dx}$

M1 A1 (RHS)

$3 \frac{x y^x}{y} \frac{dy}{dx} + 3 y^x \ln y = 3x^2 \ln y + \frac{x^3}{y} \frac{dy}{dx}$

$3 \frac{x y^x}{y} \frac{dy}{dx} - \frac{x^3}{y} \frac{dy}{dx} = 3x^2 \ln y - 3 y^x \ln y$

M1

$\frac{dy}{dx} \left(\frac{3x y^x}{y} - \frac{x^3}{y} \right) = 3x^2 \ln y - 3 y^x \ln y$

$\frac{dy}{dx} = \frac{(3x^2 \ln y - 3 y^x \ln y) \times y}{3x y^x - x^3} = \frac{3y \ln y (x^2 - y^x)}{3x y^x - x^3}$

A1

c) at $x=a$ the tangent is vertical so $\frac{dy}{dx} = \infty$

$\Rightarrow 3x y^x - x^3 = 0$

B1

$x(3y^x - x^2) = 0$

$x \neq 0$ from graph (S1)

$3y^x = x^2$

$y^x = \frac{1}{3} x^2$ (1)

$x \ln y = \ln \left(\frac{x^2}{3} \right)$ (2)

Using the eq of the curve $3y^x - x^3 \ln y = 0$ with result (1) gives:

$3 \left(\frac{1}{3} x^2 \right) - x^3 \ln y = 0$

M1

$x^2 (1 - x \ln y) = 0$ $x^2 \neq 0$

$1 = x \ln y$

Using result (2) above

$1 = \ln \left(\frac{x^2}{3} \right)$

M1 A1

$e = \frac{x^2}{3}$

M1

$3e = x^2$

$x = \pm \sqrt{3e}$

A1

$a = \sqrt{3e}$

Question 7 continued

d) at $y=b$ and $y=c$ the tangent is horizontal so $\frac{dy}{dx} = 0$

$$\Rightarrow 3y \ln y (x^2 - y^2) = 0 \quad \text{BI} \quad \ln y \neq 0 \text{ since } y \neq 1 \quad (S^+)$$

$$\begin{aligned} x^2 &= y^2 \\ \ln x^2 &= x \ln y \end{aligned}$$

Using eq of curve $3y^2 - x^3 \ln y = 0$ with this result gives $3x^2 - x^3 \ln y = 0$ M1

$$3x^2 - x^2 x \ln y = 0$$

$$3x^2 - x^2 \ln x^2 = 0 \quad \text{M1}$$

$$x^2 (3 - \ln x^2) = 0 \quad x^2 \neq 0$$

$$3 = \ln x^2$$

$$e^3 = x^2 \quad \text{A1}$$

$$\pm \sqrt{e^3} = \pm (e^3)^{1/2} = \pm e^{3/2} = x$$

from $\ln x^2 = x \ln y$ we have $3 = x \ln y$

$$\text{When } x = e^{3/2} \quad y = e^{3/e^{3/2}} = e^{3e^{-3/2}} \quad \text{M1}$$

$$\text{So } c = e^{3e^{-3/2}} \quad \text{A1}$$

$$\text{When } x = -e^{3/2} \quad y = e^{3/-e^{3/2}} = e^{-3e^{-3/2}}$$

$$\text{So } b = e^{-3e^{-3/2}} \quad \text{A1}$$



Question 7 continued

Lined area for writing the answer to Question 7.



Question 7 continued

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

(Total for Question 7 is 24 marks)

TOTAL FOR PAPER IS 100 MARKS



Pearson Edexcel Award

Tuesday 24 June 2025

Afternoon (Time: 3 hours)

Paper
reference

9811/01

**Advanced Extension Award
Mathematics**

Insert for questions 5, 6 and 7

Do not write on this insert.

Turn over ➤

P76378A

©2025 Pearson Education Ltd.
Y:1/1/1/



Pearson

5. The discrete random variable X has the probability distribution

n	2	4	6	8	10
$P(X = n)$	$\frac{1}{10} \operatorname{cosec}^2 x$	a	$\frac{1}{5} \cot^2 x$	$\frac{1}{10} \sec^2 x$	$\frac{1}{5} \tan^2 x$

where a and x are constants.

(a) Show that

$$\sin^2 2x = \frac{6}{7 - 5a} \quad (7)$$

(b) Use the result from part (a) to find a restriction on the maximum possible value of a . (2)

The random variables X_1 and X_2 are independent and each have the same distribution as X .

Given that $P(X_1 \times X_2 = 16) = a - \frac{1}{25}$

(c) determine any possible values of a , justifying your answer. (8)
(+S2)

(Total for Question 5 is 19 marks)



6. Relative to a fixed origin O , the points A , B and C have position vectors

$$\vec{OA} = 10\mathbf{i} + 5\mathbf{j} + 4\mathbf{k} \quad \vec{OB} = -4\mathbf{i} + 7\mathbf{j} - \mathbf{k} \quad \vec{OC} = -2\mathbf{i} + 17\mathbf{j} + 10\mathbf{k}$$

- (a) Show that triangle ABC is isosceles.

(2)

The circle S lies inside triangle ABC such that each side of the triangle is a tangent to S .

The centre, X , of S lies on the angle bisector of each vertex of the triangle.

The point D lies on AC such that $|\vec{AD}| = |\vec{AB}|$

- (b) Find the position vector of D .

(2)

- (c) Find the position vector of the midpoint of BD .

(2)

- (d) Hence show that the position of X is given by

$$\vec{OX} = 10\mathbf{i} + 5\mathbf{j} + 4\mathbf{k} + p(-2\mathbf{i} + \mathbf{j})$$

where p is a scalar constant.

(4)

- (e) Use the angle bisector of vertex B to find another expression for $\vec{OX}$ in terms of a scalar constant q .

(3)

- (f) By equating the $\mathbf{k}$ components of the two expressions for $\vec{OX}$, determine the value of q .

(2)

- (g) Find

- (i) the position vector of the centre of S ,

(2)

- (ii) the radius of S .

(4)

(+S2)

(Total for Question 6 is 23 marks)

7. Given $u = y^x$

(a) by taking $\ln$ of both sides, or otherwise, determine $\frac{du}{dx}$ in terms of x, y and $\frac{dy}{dx}$ only. (5)

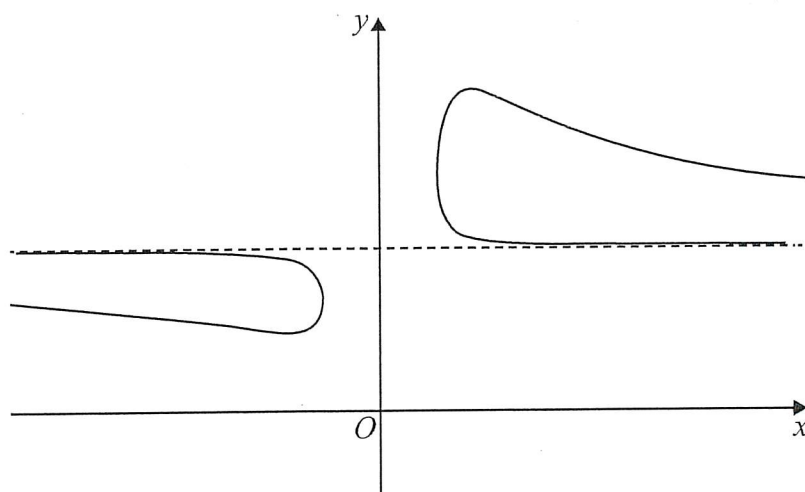


Figure 1

Figure 1 shows a sketch of the curve given by the equation

$$3y^x - x^3 \ln y = 0$$

The curve has a horizontal asymptote $y = 1$, and is defined for

- $x \leq -a, \quad x \geq a$
- $b \leq y \leq c \quad y \neq 1$

where a, b and c are positive constants.

(b) Determine $\frac{dy}{dx}$ for this curve. (4)

(c) Find the value of a . (6)

(d) Find the value of b and the value of c . (7)
(+S2)

(Total for Question 7 is 24 marks)